Template Journal of Computing, Data, and Exploration

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Abstract - This document is the article writing format for the Journal of Computing, Data, and Exploration (CODEX). Authors must follow all instructions in this document to be published in the CODEX. Abstract is a summary of the article. The abstract contains a summary of the introduction, a summary of the methods, a summary of the results and discussion, and a summary of the conclusions. The number of words in the abstract ranges from 200 - 300 words. The abstract is written in **English** using **font size 10pt spacing single before 12pt after 12pt italic** style. Keywords are scientific terms that are often encountered in the field of science related to the article, and can be used as a search key to find articles. Use 3-5 words or phrases that are key to the submitted article. *You should not use symbols, mathematical equations in the title and abstract.

***Keywords*:** keywords; are written; alphabetically; with a maximum; of 5 words

1 INTRODUCTION

Articles should be submitted to the CODEX using this template.

When we started STA258, it became clear pretty quickly that understanding the theory was only one part of the challenge. What really tested us was figuring out how to apply those ideas to real data and make sense of what the results actually meant. Concepts like confidence intervals or regression equations seemed straightforward in class, but when it came time to use them, things felt less clear. We kept thinking—what if there was something more hands-on that could help connect the dots? That thought is what eventually led to this project. We wanted to create something we wished we had while learning—an interactive e-book that doesn't just explain the material but gives students the chance to explore it in a way that's practical, visual, and engaging. From the start, we pictured a resource made for students like us: maybe you're still getting comfortable with R, or maybe you're just looking for a clearer explanation than what's in the slides. Either way, we wanted to offer explanations, examples, and interactive components that feel more approachable and less intimidating. To make it all work, we built the project using tools like R Markdown, GitHub, and Bookdown. This setup let us collaborate smoothly, keep everything organized, and make the content reproducible and open-source for others who want to use or expand it later on. In the end, this e-book reflects everything we learned—not just about statistics, but about building something useful from scratch. While it's designed to support future STA258 students, we hope it ends up being helpful to anyone who's trying to get a better grasp of applied statistics.

2 RESEARCH METHODOLOGY

The method contains a description of how to carry out the research. It describes how to obtain data, the algorithm or formula used in the research or how to process the data, and how to evaluate/assess the research results. Common methods do not need to be written in detail, but simply refer to the reference book. Research procedures should be written in the form of news sentences, not command sentences.

2.1 Page Style

The easiest way to fulfill the formatting requirements is to use this document as a *template*. *Author* can directly write on this *template* file. Articles are written in A4 paper format with a right-left-bottom border of 2 cm and an upper border of 3.25 cm. The article format uses 1 column.

3 RESULTS AND DISCUSSION

Research results in the form of data or numbers are presented in tables or graphs. If the research is carried out application/software development, some important *screenshots* can be presented. Each table, graph or figure must be referred to in the text/paragraph.

The discussion section provides *insights* into the data obtained in the research. This section can present tables or graphs that are the result of data processing (not just raw data). The author is required to explain the findings obtained in the research, accompanied by clear evidence. This section can contain reviews that compare the results obtained in this study with the results obtained in previous studies.

1. Writing Format

Paragraphs should be organized and consistent. Pay attention to the spelling and formatting of foreign terms. All paragraphs must be right-aligned and left-aligned. The entire document should be written in Times New Roman font with *single* spacing. Other fonts may be used if there is a specific purpose. The title consists of a maximum of 10 words.

2. *Author*

Authors should indicate their names. To avoid confusion, the last name of each *author* should be written at the end, not abbreviated and marked with a comma (e.g. Rizky Prabowo becomes Prabowo, Rizky). Each affiliation must include, at a minimum, the name of the company and the name of the author's country of residence (e.g. University of Lampung, Indonesia). An email address is required for the corresponding author.

3. Math Formulation

Equations must be written using the LaTex. Writing symbol descriptions in equations is made in descriptive paragraphs, not list items as in book writing. Equations must be typed with an indent of 1.27 pt and numbered sequentially starting with the number (1) on the right. *Author* can use *ref_rumus style* for formula writing rules.

$$\int_0^\infty e^{-x^2} dx = \frac{\sqrt{\pi}}{2}\,(1) \tag{3.1}$$

4. Table and Figure/Graphic Writing

All figures and tables must be centred and numbered consecutively. All text within figures and tables must be clearly legible, no *blurring*. Each figure and table must be referred to in the text, the way of referring should not use location (e.g. below, above, following, etc.). Writing tables, figures, graphs, and equations must have high resolution (min 300 dpi). Avoid cropping images or *screenshots* for writing tables and equations.

Figure 1: UTM logo

The *author* can use the image *style* for image writing rules while for table writing, the *author* can use the Table *style*. Tables are made with horizontal lines without using vertical lines. Tables must not be cut off on other pages.

No.	Segments	Length (km)	Elevation (meters)
1	A-B	25	30
2	B-C	75.15	10
3	C-D	44.75	50
4	D-E	72.5	10
5	E-F	21.25	10

Table 1: Summary of physical parameters

5. Bibliography Writing

References in the text must be cited by writing the author's last name and its sequence number in the reference. The citation style used in Journal of Computing, Data, and Exploration is IEEE [1]. Bibliography writing also uses IEEE style. Citations are sorted by their appearance in the text and use Arabic numerals in square brackets [2]. It is highly recommended to use citation management applications such as Mendeley or Reference Manager when writing articles [3], [4], [5] so that citation management becomes easy. Writing a list of references using single *spacing* using *font size 11pt, spacing before 0 after 6pt*.

4 CONCLUSIONS

The Conclusion section summarizes the reviews written in the Results and Discussion sections. The Conclusion section is written in paragraph form, no need to use numbers. The paper will not be reformatted, so please stick to the instructions given above. Otherwise, it will be returned to the *author*. Articles should be returned back via e-mail to be processed.

NOMENCLATURES

A = Amplitude
C_d = drag coefficient
f_e = linearization coefficient
K_i = modification factor

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